Team Fire

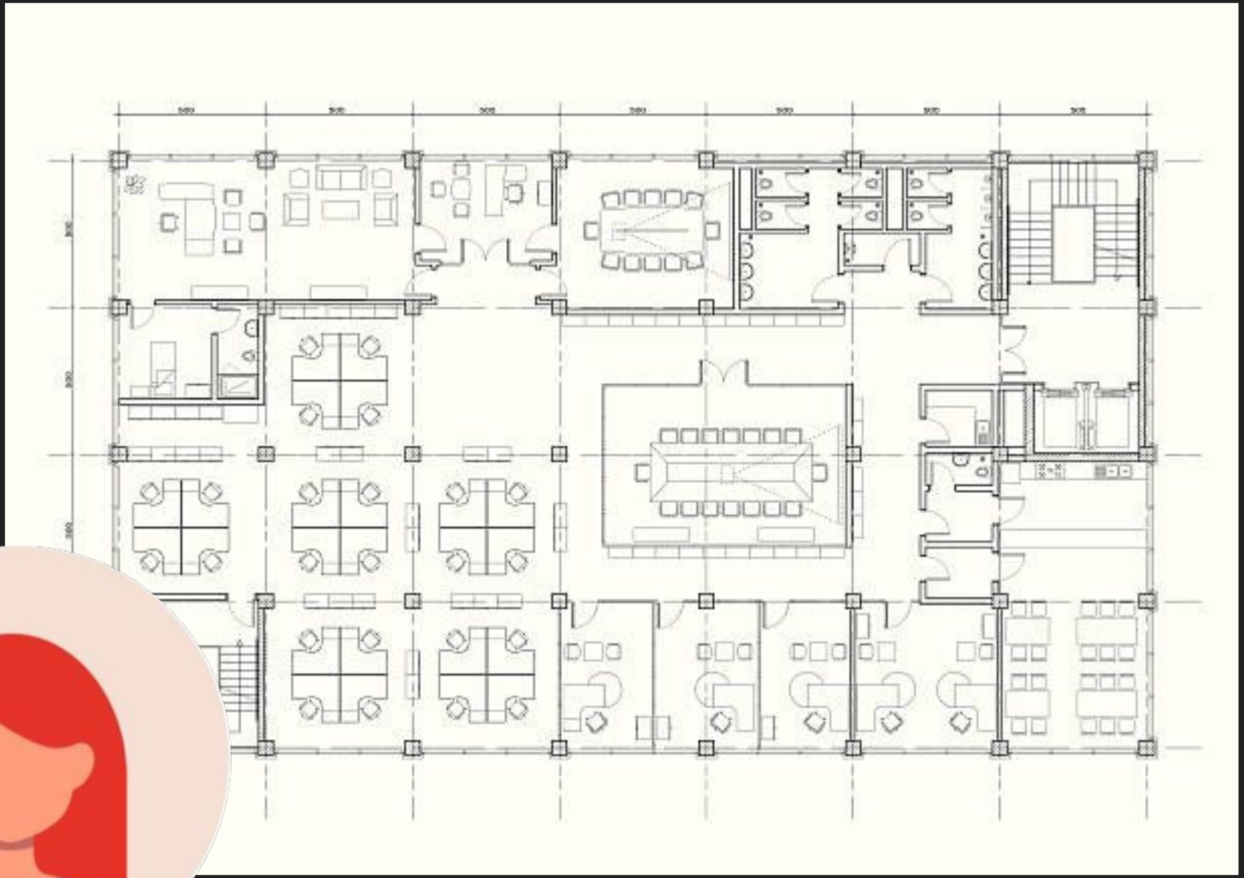
AI Powered Compute API & Plugin for Fire Compliance Checking & Solving

TEAM 🔥

- **Atenn Neoh**: Architectural Developer
- **Yvonne Zhang**: Fire Engineering Subject Matter Specialist
- **Bob YX Lee**: Systems Integration, Full Stack Development
- **Tang Minjing**: BIM Specialist
- **Evangelina Ong**: BIM Specialist
- **Yuji Fujinami**: UI/UX, Frontend Development, Machine Learning Implementation

How Traditional Fire Compliance Checks are done

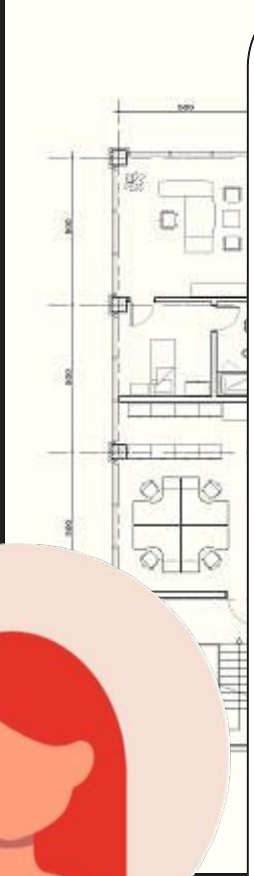
Architect



Example of an
office floor plan

Fire Extinguisher
Placement

How Traditional Fire Compliance Checks are done



Architect

6.1 PORTABLE EXTINGUISHERS

6.1.1 General

a. Portable fire extinguishers, where required, shall be constructed in accordance with *SS EN 3*.

b. All portable fire extinguishers, where required to be provided, shall be charged, tested, maintained and properly tagged in accordance with *SS 578*.

6.1.2 Provision

a. Fire extinguishers shall be provided in all buildings except the following: E

(1) PG I buildings;

(2) residential floors of PG II buildings;

(3) car parking areas in standalone car parks or mixed-use residential buildings;

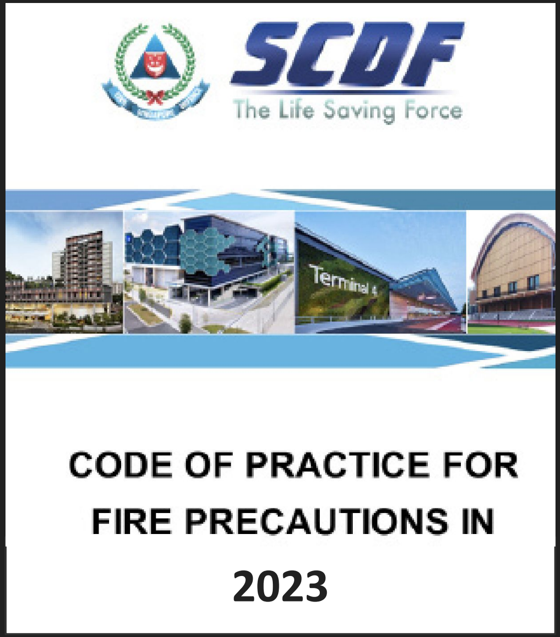
(4) roof level of single storey buildings with roof height not more than 12m or inaccessible pitched roof up to 24m from grade level used solely for roof-mounted PV installations in accordance with *Cl.10.2.1b.(1)*; and

(5) roof level of an external/ open-sided overhead bridge/ shed/ linkway/ walkway with clear width less than 6m, roof height not more than 12m and used solely for roof-mounted PV installations in accordance with *Cl.10.2.1b.(1)*.

Effective Date: 01 Mar 2019

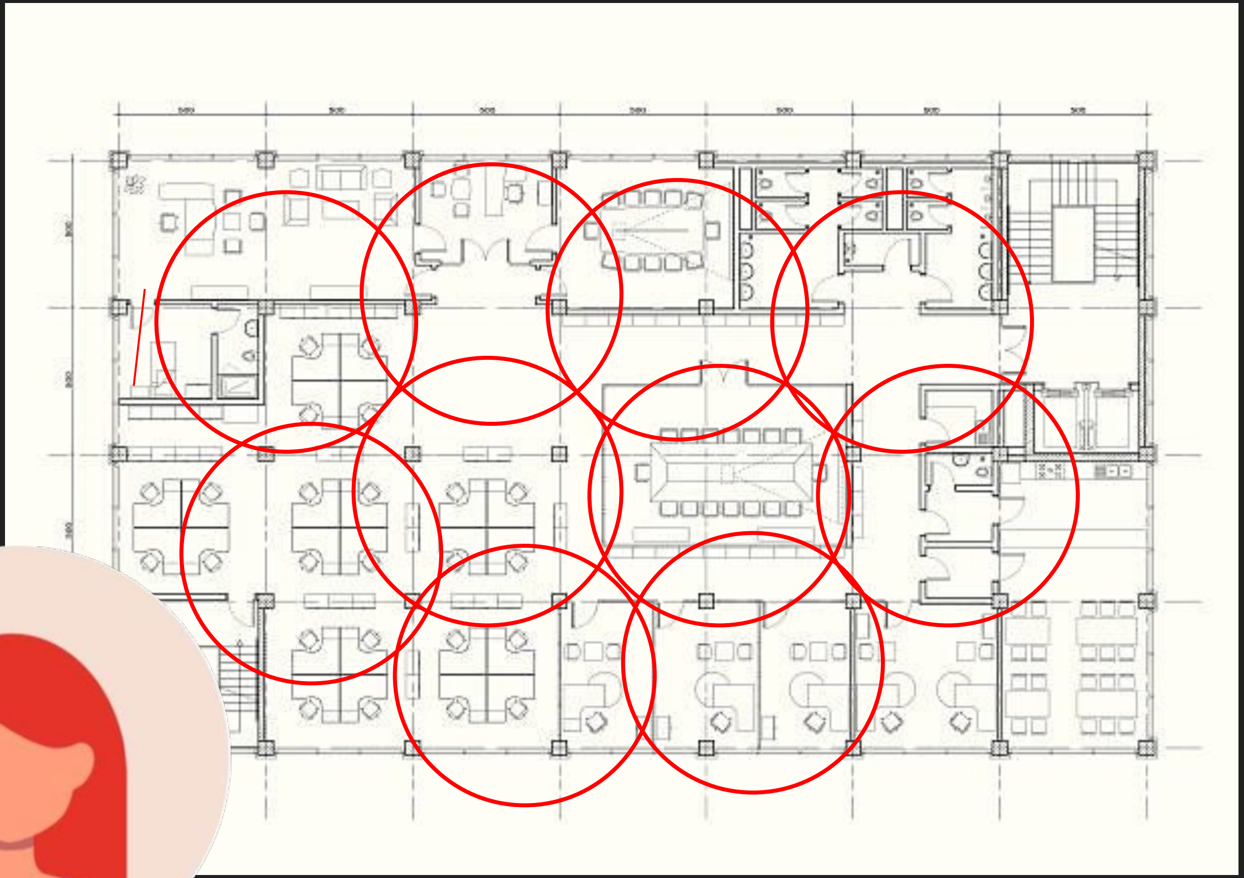
b. Where the roof level is a non-habitable floor, fire extinguishers shall be provided to cover the M&E plants/ equipment.

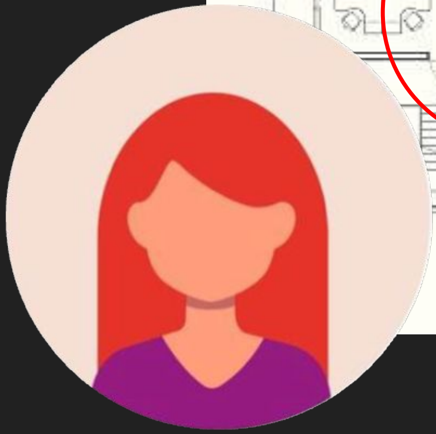
6.1.3 Type, size and siting E



Refer to guidelines

How Traditional Fire Compliance Checks are done





Architect

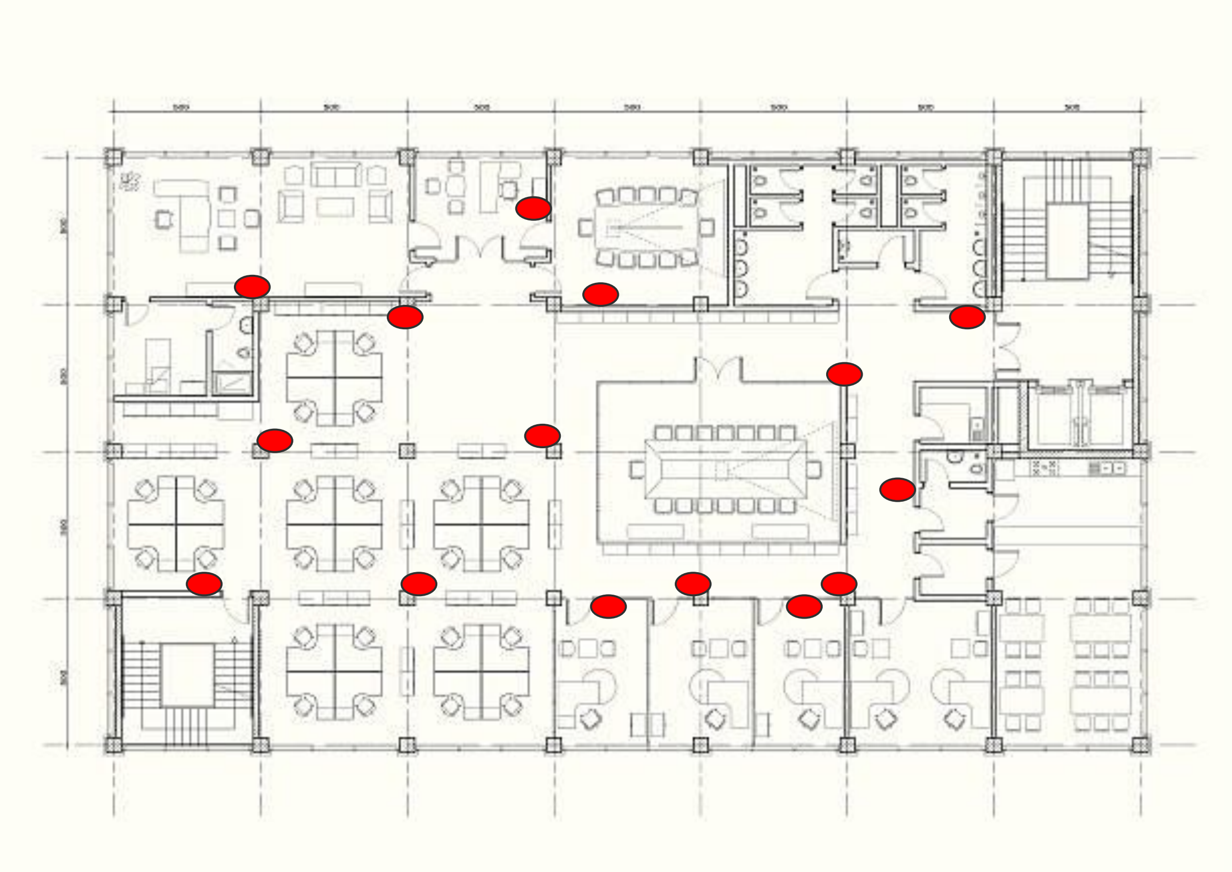
1. Review Fire Extinguisher Placements in order of priority
2. Ensure placement of every fire extinguishers cover 15m radius
3. Ensure actual walking path to next fire extinguisher is 15m

Total Time Taken: ~1-2 hours or more

How Traditional Fire Compliance Checks are done

XX no. of fire extinguishers

Suggested placements are shown

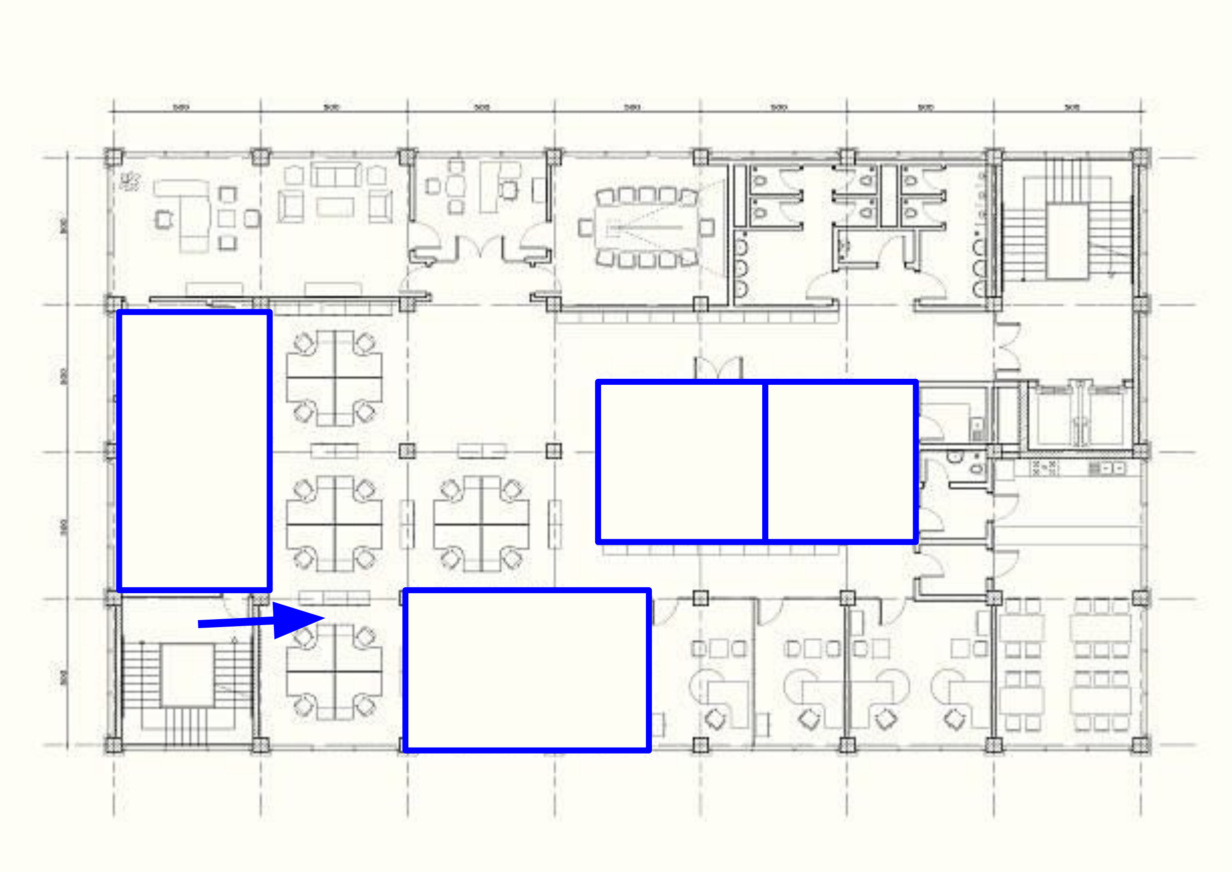


Architect

Are you 100% certain?
I will allocate this amount for the extinguishers

Project Manager of a Developer

How Traditional Fire Compliance Checks are done

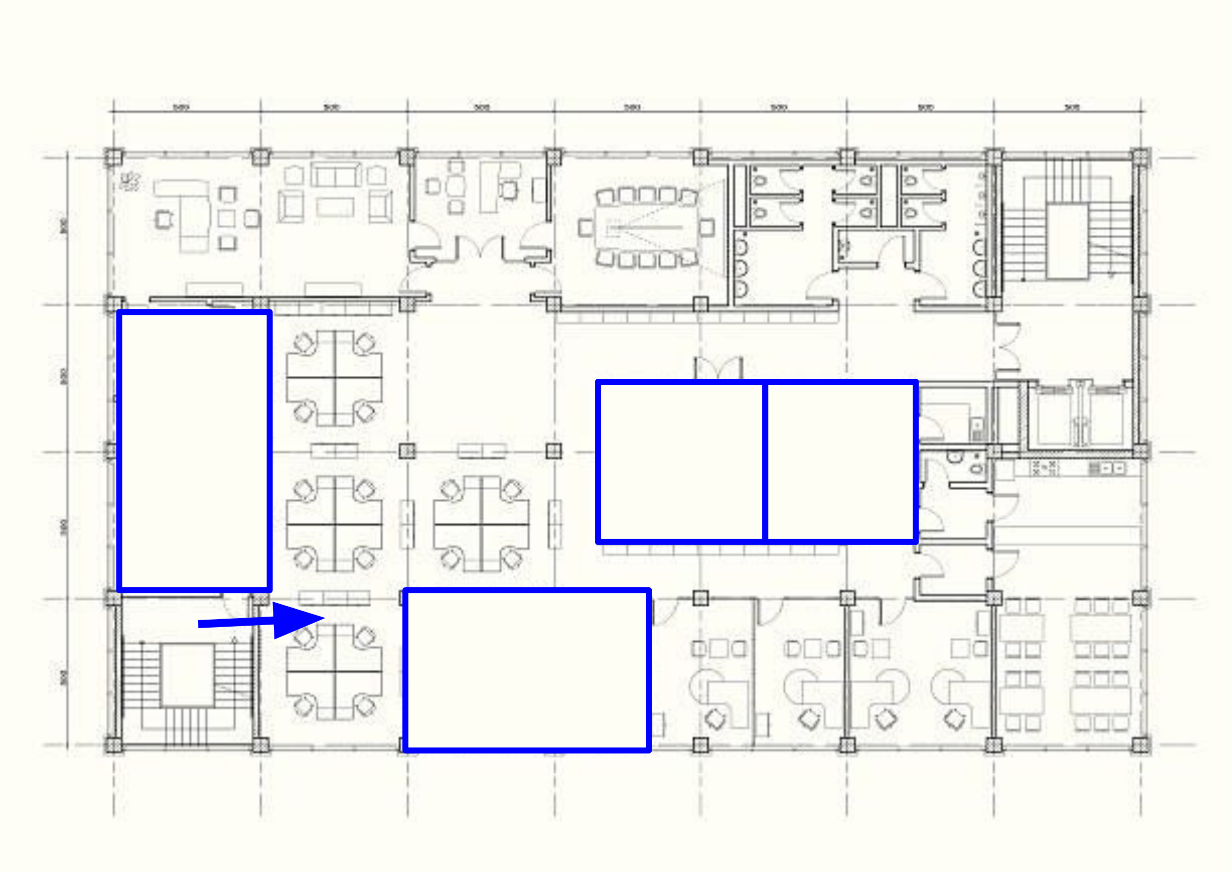


Architect

I would like to add more rooms.
Change the position of the doors.
Change the usage of the rooms - Electrical

Project Manager of a Developer

How Traditional Fire Compliance Checks are done

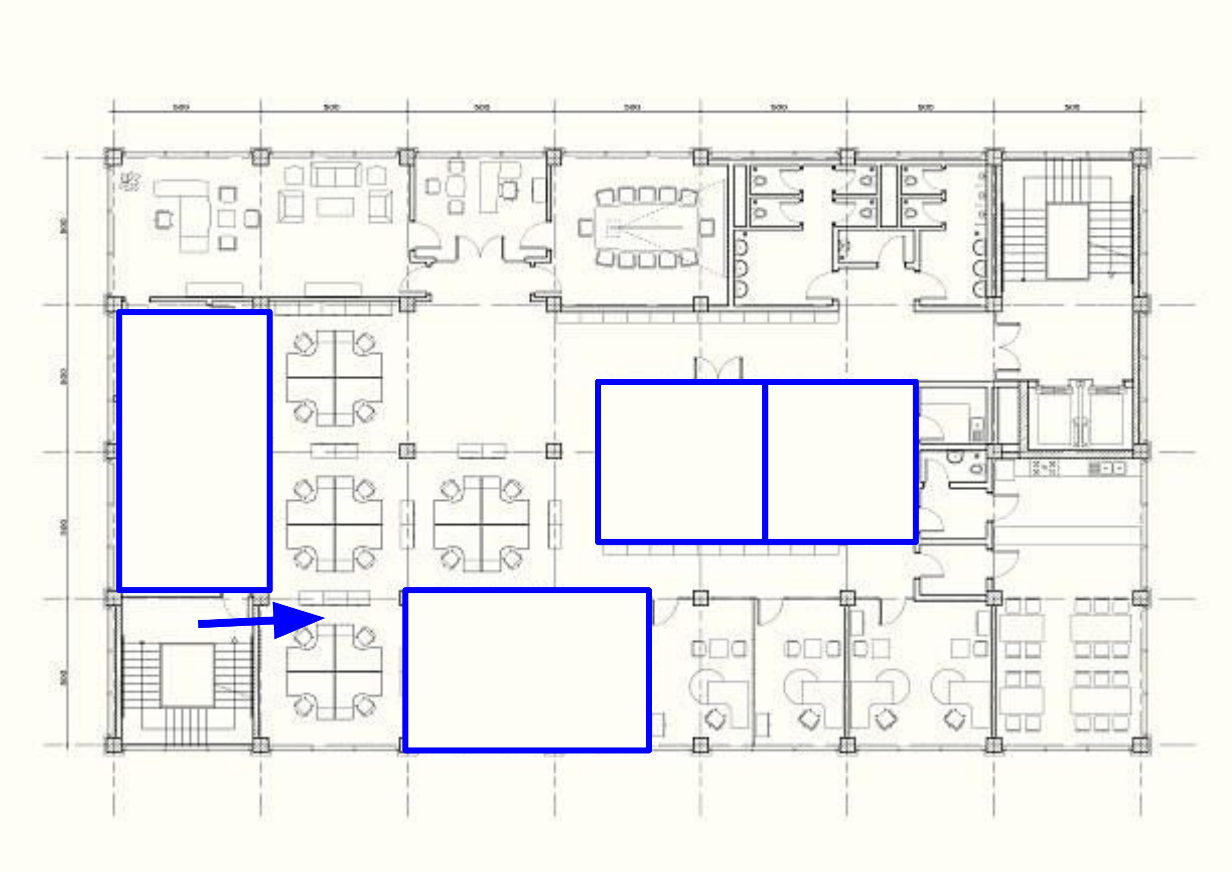


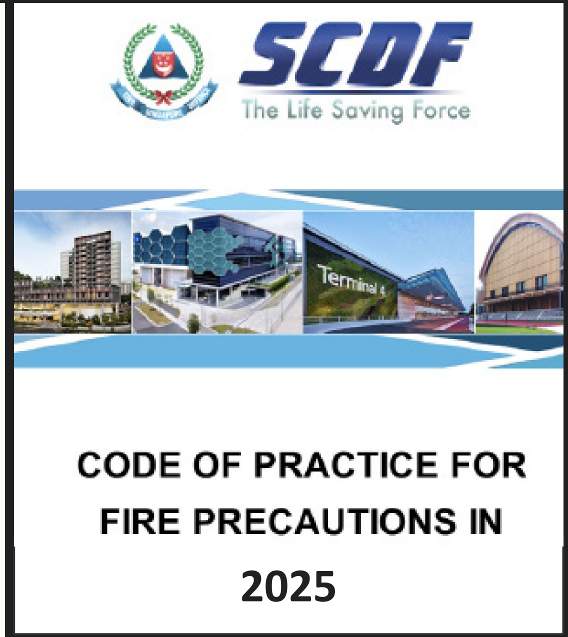
Can I have the final amount of items needed to cost for FSSD first? I need to report to my team tomorrow

Architect

Project Manager of a Developer

How Traditional Fire Compliance Checks are done





Guidelines revision!

Architect

Pain Points

- Inefficient and difficult to give an immediate evaluation
- Extremely Repetitive and Laborious checking and measuring
- Prone to human error due to lack of time or guidelines changes

Architect

Time spent for checks CANNOT BE COSTED

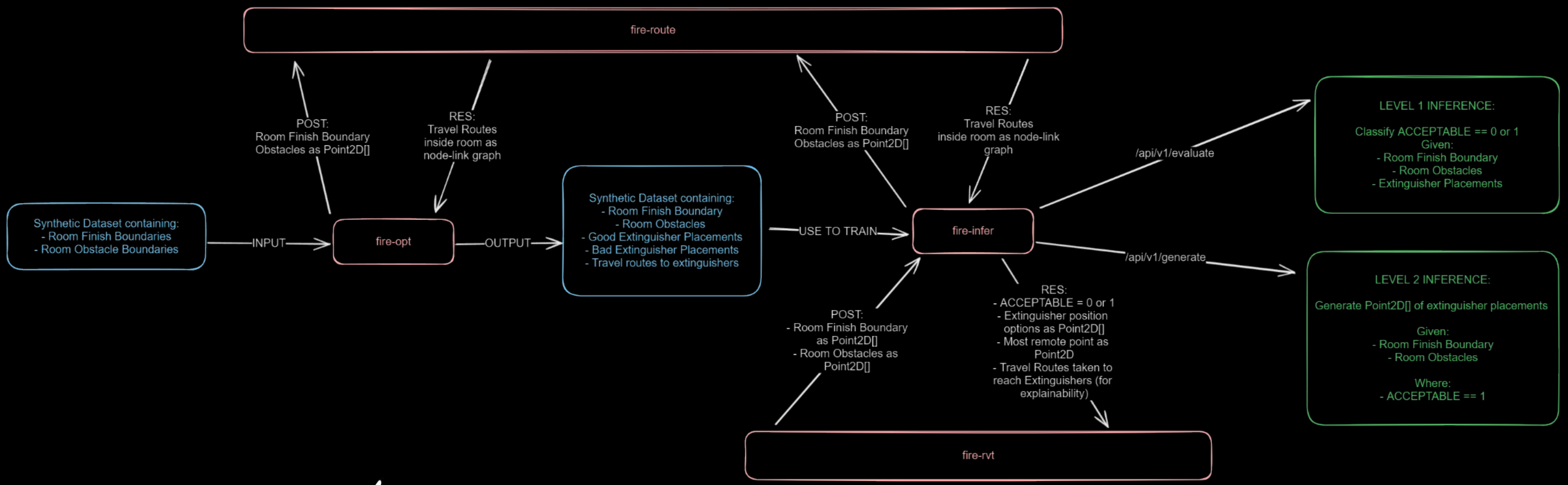
Solution

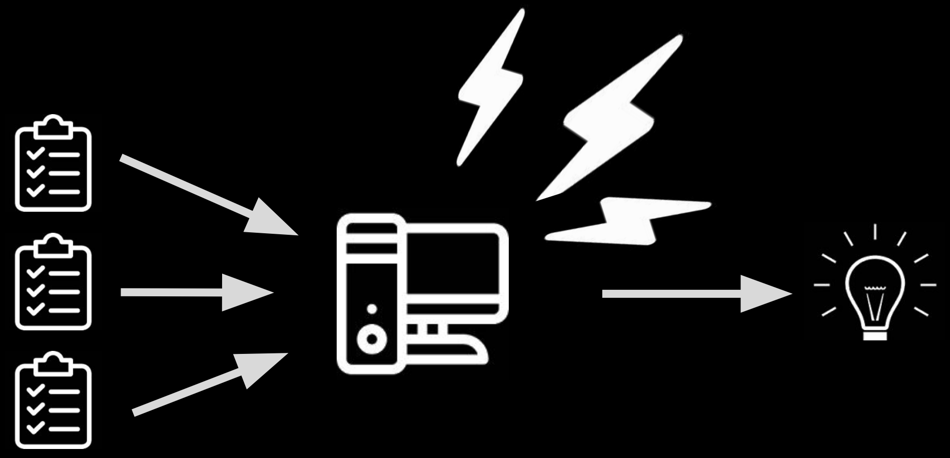
Our Team intended to create an AI that is able to identify the locations of the FSSD items, specifically Fire Extinguishers in our proposal, to be placed on plans. The checks will also serve to highlight any errors or discrepancies of non-compliances within seconds.

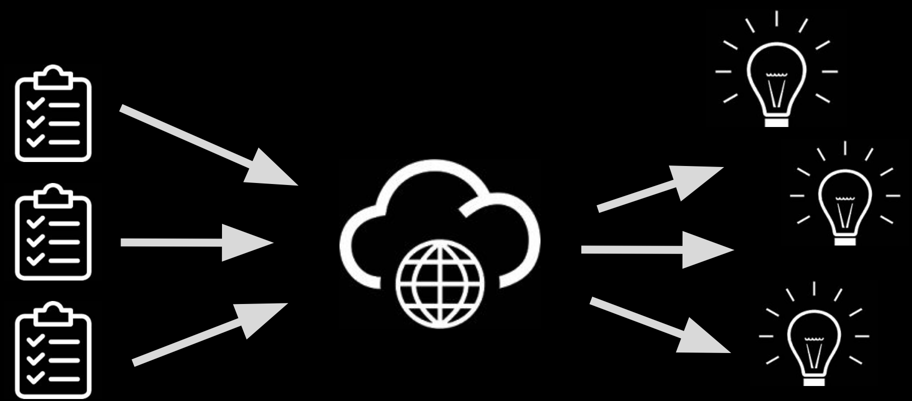
Tool

Workloads for 🧯 AI Inference

1. Check coverage with 15m radius circle around 🧯
2. Check actual walking path to determine if it is indeed within 15m
3. If not within 15m, adjust placement - possibly adding new 🧯


fire-route
POST:
Room Finish Boundary
Obstacles as Point2D[]
RES:
Travel Routes
inside room as
node-link graph
Synthetic Dataset containing:
- Room Finish Boundaries
- Room Obstacle Boundaries
INPUT
fire-opt
OUTPUT
Synthetic Dataset containing:
- Room Finish Boundary
- Room Obstacles
- Good Extinguisher Placements
- Bad Extinguisher Placements
- Travel routes to extinguishers
USE TO TRAIN
POST:
Room Finish Boundary
Obstacles as Point2D[]
RES:
Travel Routes
inside room as node-link
graph
fire-infer
/api/v1/evaluate
/api/v1/generate
LEVEL 1 INFERENCE:
Classify ACCEPTABLE == 0 or 1
Given:
- Room Finish Boundary
- Room Obstacles
- Extinguisher Placements
LEVEL 2 INFERENCE:
Generate Point2D[] of extinguisher placements
Given:
- Room Finish Boundary
- Room Obstacles
Where:
- ACCEPTABLE == 1
POST:
- Room Finish Boundary
as Point2D[]
- Room Obstacles as
Point2D[]
RES:
- ACCEPTABLE = 0 or 1
- Extinguisher position
options as Point2D[]
- Most remote point as
Point2D
- Travel Routes taken to
reach Extinguishers (for
explainability)
fire-rvt

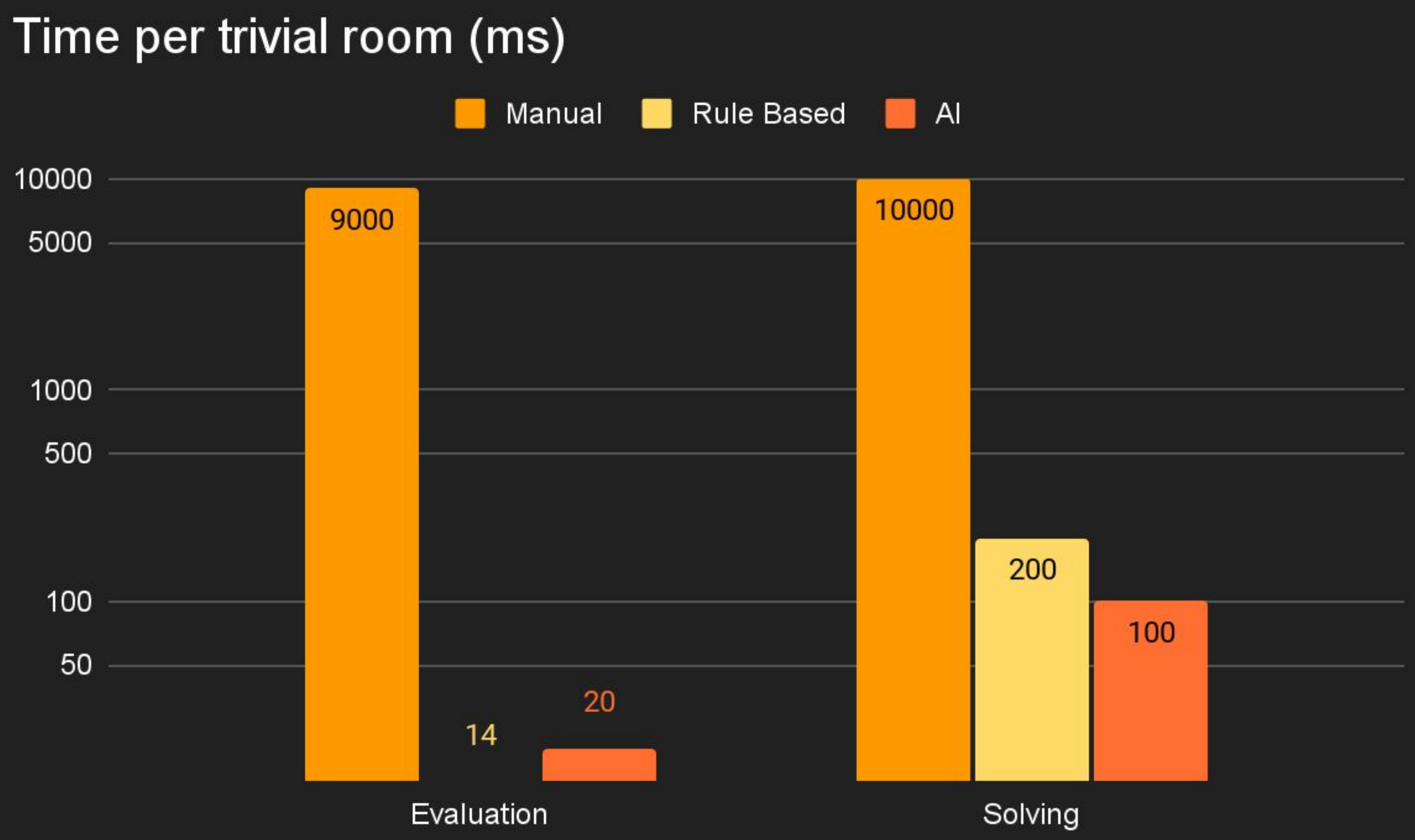




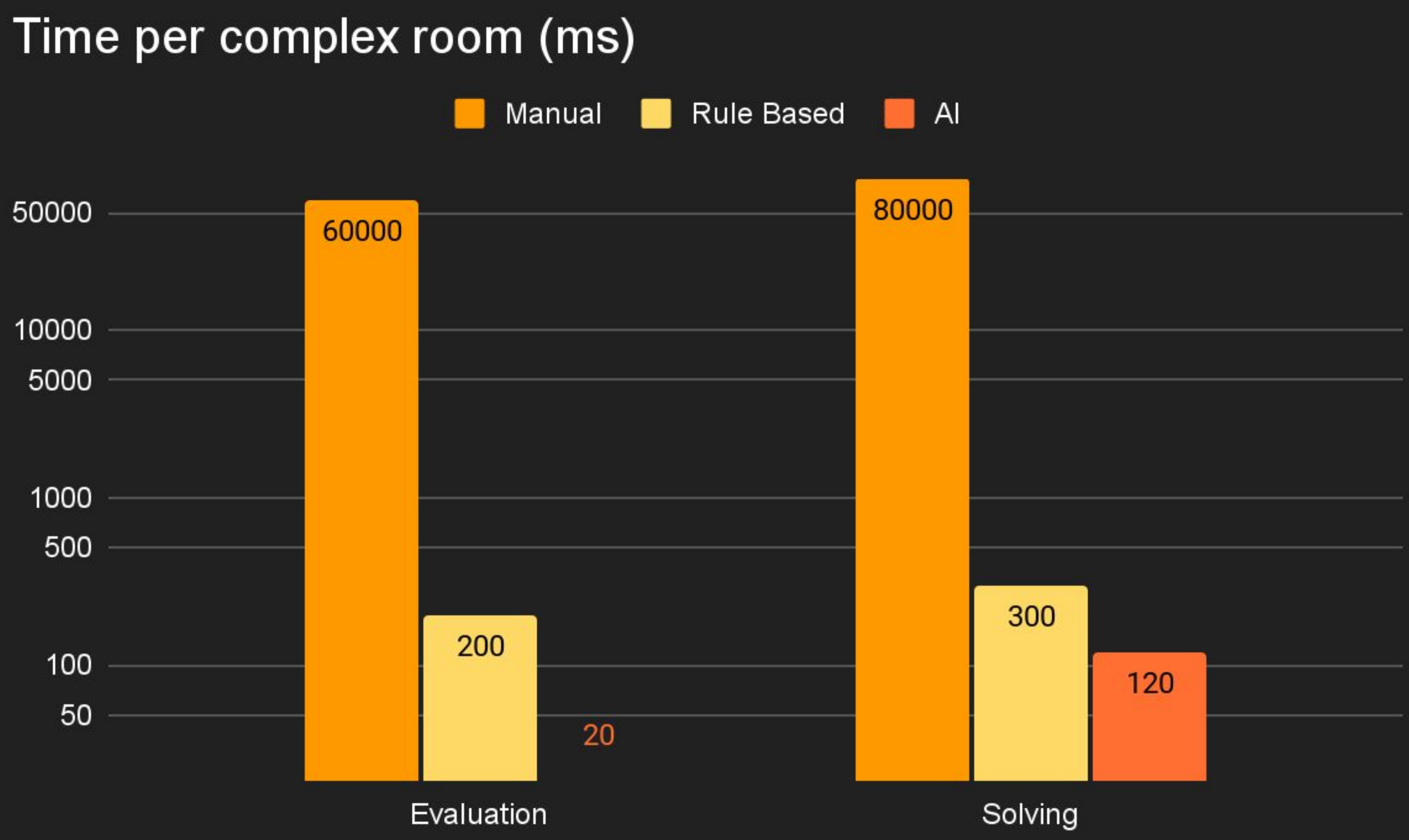
Solution Flow

- User loads 🧯 locations & floor plans from Revit using a plugin
- Plugin opens a web page to render the draft extinguisher options
- User selects the floor & room to solve for
- User presses “Check” to check for compliance
- Or, User presses “Suggest” to request inference for the ideal 🧯 layout
- Data is sent to an API server to compute the compliance / inference and returned for the web page to preview

Performance for trivial small scale problems

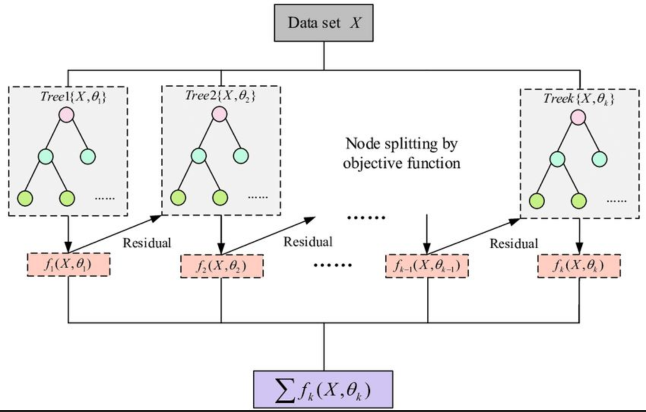


Performance for **complex** large scale problems



AI Architecture - Compliance Predictor using ML

- Trained and evaluated several AI models based on a **synthetic dataset** of **10,000 data points**, to predict whether a plan with fire-extinguisher placements is compliant with the Fire Code and Singapore Standard.
- Feature engineering performed on dataset to create features that would improve the predictive performance of the AI models - i.e. Room Area, Average Distances of the Extinguishers to the Centre of the Room, etc.
- Concluded that XGBoost Classifier performed the best among the AI models tested.
- XGBoost Classifier - ensemble method utilizing decision trees as base learners.
- High Accuracy and F1 Scores achieved.



Pipeline

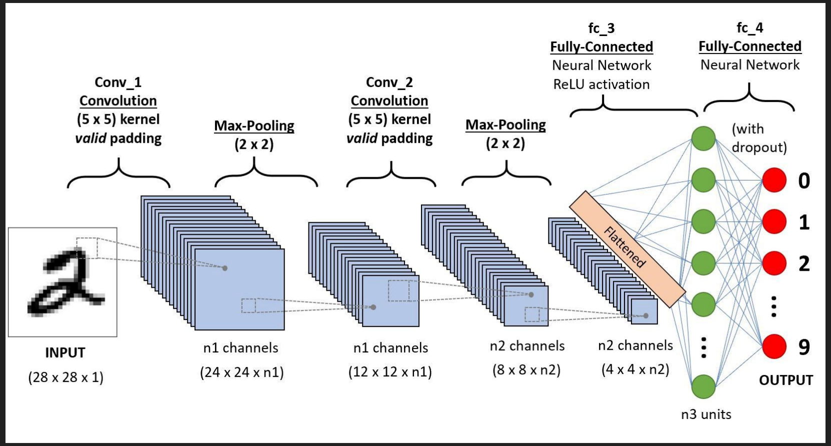
```
# Target variable
y = df['comply']

# Train test split
X_train, X_test, y_train, y_test = train_test_split(df.drop(columns=['comply']), y, test_size=0.2, random_state=42)

# Train and evaluate models
results = train_and_evaluate_models(X_train, X_test, y_train, y_test)

# Convert results into dataframe
results_df = pd.DataFrame(np.array([["SVM Classifier",results[0][1],results[0][2],results[0][3],results[0][4]],["XGB

print(results_df)
print(results)
```

```
            Model Accuracy Score  F1 Score ROC AUC Score  \
0  SVM Classifier          0.938  0.962447      0.859539
1  XGB Classifier         0.9425  0.964992      0.874834
```

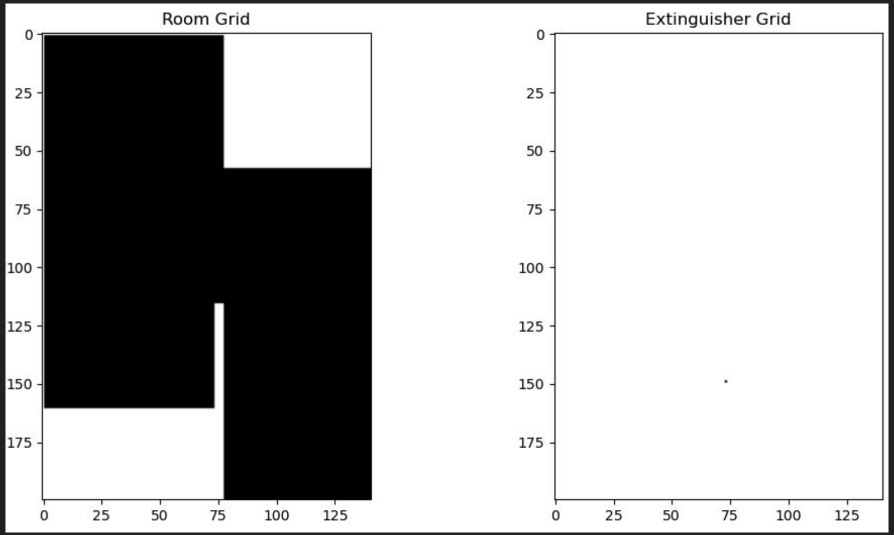
AI Architecture - Placement Inferencer

Convolutional Neural Network (CNN)

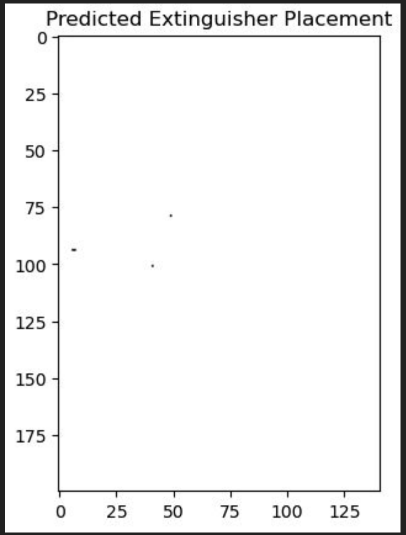
- Trained a CNN using the same **synthetic dataset** of **10,000 data points**, to predict and output compliant fire-extinguisher placements given a plan.
- Transformed and generated binary grids for the room plan and extinguisher placement coordinates.
- Final trained CNN model performed badly on test set - i.e. giving the same output for test data points.
- Likely suffering from a sparsity issue as binary images are mostly filled with zeroes, greatly outnumbering the non-zero (extinguisher) pixels.
- Further improvements include data augmentation and regularization.



Original



Prediction



Conclusion

Innovation

- Current market tools are not efficient enough to evaluate and suggest compliance methods using AI.
- With Project Fire, we've demonstrated the minimal implementation of a highly efficient AI tech stack for:
    - Rapid compliance verification
    - Rapid optimisation of the placement of fire implements

Conclusion

Innovation

- New highly efficient AI tech stack for:
  - Rapid compliance verification
  - Rapid optimisation of the placement of fire implements

Impact

- As demonstrated with our tool, it is able to verify and predict with high accuracy the placement of FSSD items adhering to compliance guidelines.
- Project Fire allows the user to maxmise his time by focusing on design matters, leaving technicalities to AI
  - More accurate placement
  - 90% of time savings compared to manual
  - Less taxing on manpower and cost of checking
  - Overall massive productivity gains

Conclusion

Innovation

- New highly efficient AI tech stack for:
    - Rapid compliance verification
    - Rapid optimisation of the placement of fire implements

Impact

- Our AI tools allows for:
    - More accurate placement
    - 90% of time savings compared to manual
    - Less taxing on manpower and cost of checking
    - Overall massive productivity gains

Scalable

- The same paradigm and solution architecture may be applied for other related applications e.g., Fire Hosereel, Fire Hydrant and may even be extended to other areas such as Lighting and Furniture Placement.

Conclusion

Scalability in terms of Computational Power

- The solution is highly scalable as it offloads compute onto a server, allowing the practice to scale compute on cloud or on premises using the same solution as and when needed (i.e., users may use low-spec computer to harness its capabilities)

Future Plans

- Fire Extinguisher Placement - Solve for some obstacles in each room <<
- Hydrant Placement - Not as many obstacles to consider, too simple
- Fire Hose Reel Placement - Solve for plan and obstacles in multiple adjacent rooms, a bit too complex for hackathon
- Riser Placement - Solve for plan across multiple floors, very complex, etc

FSSD

Future Plans

Our program serves to automate checks extending to all Authorities and inform the Architect/ Client immediately for any non-compliances.

The burden of checks is thus, entrusted to AI which is more efficient and accurate.

FSSD URA BCA Nparks NEA PUB

Future Plans

Authorities

FSSD URA BCA Nparks NEA PUB

Future Plans

M&E

Structural

Architectural plan layout

Authorities

FSSD URA BCA Nparks NEA PUB

Thank You

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https://github.com/boblyx/fire